

Prathyush Sajith

+1-312-684-6665 | psaji@uic.edu | prathyushsajith17@gmail.com | [Linkedin](#) | [Portfolio](#)

Chicago, Illinois - 60612

EDUCATION

• University of Illinois at Chicago

Aug 2026 (expected)

MS Thesis in Computer Engineering

◦ Advisor: Prof. [Ahmet Enis Cetin](#)

◦ Selected Courses: Convex Optimization, Neural Networks, Image Analysis and CV, Pattern Recognition.

• SRM Institute of Science and Technology, Ramapuram

May 2024

B.Tech in Computer Science Engineering

◦ Grade: 9.20/10.00 (First class with Distinction)

EXPERIENCE

Graduate Research Assistant – Machine Learning & Medical Imaging

Sep 2025 – Present

[Tools used: Python - Pandas, PyTorch, Scikit-learn, OpenCV]

- Developed end-to-end data preprocessing and cleaning pipelines for large-scale 3D/4D two-photon microscopy datasets, including intensity normalization, noise reduction, and patch extraction for volumetric learning tasks.
- Developing deep learning models using Pytorch for automated segmentation of blood-brain barrier tight junction structures (Claudin-5), replacing manual image quantification with reproducible analysis.
- Worked with sensitive patient data with HIPAA guidelines, while collaborating with professors and postdoc researchers to align ML objectives to interpret results in a biological context.

MS THESIS

Improving Efficiency of Stereo Depth Estimation using Walsh-Hadamard Transforms

May 2025 – Present

[Tools used: PyTorch, TorchVision, OpenCV, CUDA, cuDNN, PIL, Matplotlib]

- Designed and extended a **Stereo Transformer (STTR)** based depth estimation pipeline, leveraging attention-driven correspondence modeling along epipolar lines for long-range disparity estimation.
- Improved computational efficiency by replacing standard Conv2D layers with a Walsh-Hadamard Transform-based convolution (WHTConv2D) **by 9%**, reducing complexity while preserving representational capacity.
- Implemented a **Log-Disparity Loss formulation** to emphasize supervision on far-range (>100 m) depth regions, improving stability and accuracy for small-disparity predictions.

SELECTED PROJECTS

(1) Stereo Depth Estimation using PSMNet ([git](#))

Aug – Sep 2025

[Tools used: PyTorch, TorchVision, cuDNN]

Adapted a pyramid matching CNN for depth estimation by fine-tuning on a custom 4K stereo dataset. Engineered a data pipeline to resolve non-standard image dimensions and disparity channel mismatches. Implemented a final disparity-to-distance conversion module to enable real-world distance measurement from the model's predictions.

(2) Custom built GPT-Based Large Language Model ([git](#))

Jan – Feb 2025

[Tools used: Pytorch, Hugging Face Transformers, NumPy, Pandas]

Built a 124M parameter GPT-based LLM using Pytorch. Designed an efficient tokenizer, created embeddings with 768-1280 dimensions, and implemented multi-head self-attention and rotary embeddings. Pretrained the model on up to 1024 tokens per context window and achieved significant loss reduction. Fine-tuned the model to generate coherent text outputs, by data preprocessing and optimization.

(3) Lane Detection using UNet implemented as Discrete Wavelet Transform ([git](#))

Nov – Dec 2024

[Tools used: Python, Keras, TensorFlow-GPU, OpenCV // Architecture: Discrete Wavelet Transform, U-Net]

Developed a deep learning model for lane detection that integrates UNet architecture as a Discrete Wavelet Transform (DWT). It enhances detection accuracy, while reducing the aliasing effects, that's usually found during the upsampling process. It is implemented using GPU kernel programming in CUDA.

PUBLICATIONS

(1) Crowd analysis and tracking for individual rescue operation using machine learning. IJRASET 2024 | [ResearchGate](#)

SKILLS

- **Languages:** Python, SQL, (familiar with C and C++).
- **Deep Learning:** Tensorflow & Keras, Pytorch, Scikit-learn, Numpy, LangChain, Pandas, Matplotlib, OpenCV.
- **Analysis:** Matlab, Tableau, Microsoft Power BI, Matplotlib and Seaborn, Excel, Qualtrics, Forms, Data Cleaning, EDA.